

DNA Barcoding of Duckweed Strains

Mortimer Lab v1.1

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[Protocol originally developed by Daniel McKay]

DNA barcoding is to be used to distinguish and label the collected Australian duckweed isolates with species. Barcoding follows the protocol in Borisjuk 2014 (<https://doi.org/10.1111/plb.12229>).

Identification is by amplifying up two non-coding regions of plastid DNA, sequencing the regions and BLASTing to identify which species the region's most closely align to.

-gDNA extraction performed using basic protocol below. (Fastest and simplest. Worked first time for all lines so far. Some bacterial and for line 7, algal, contamination was present)

-PCR performed using dreamtaq in initial experiments but should probably use proofreading enzyme in future) using **atp** and **psb** primer pairs.

-PCR reaction run on on gel. Bands cut from gel and purified using gel purification kit.

-Sequencing performed by AGRF Sanger sequencing on atp first. psb will only be used for identification of species which cannot be distinguished by the atp region alone. For each line, 2 samples will be sent, one to be sequenced by the forward and one by the reverse primer. The results are stitched to give a full read (although we've found not always necessary to have both, we still do it).

Genomic DNA Extraction protocol:

- Collect several Duckweed fronds (6-10ish) in a 1.5mL tube
- Homogenise in tube with a clean blue pestle
- Add 10uL of 0.5M NaOH
- Centrifuge at 15,000 rpm for 30 seconds at RT
- Transfer 5uL of supernatant into a clean 1.5 mL tube containing 50uL of 100mM Tris-HCl (pH 8.0)
- Store at -20oC

PCR performed using primer pairs:

	fwd		rev	
atp:	atpF	ACTCGCACACACTCCCTTTCC	atpH	GCTTTTATGGAAGCTTTAACAAT
psb:	psbK	TTAGCATTGTGTTGGCAAG	psbI	AAAGTTTGAGAGTAAGCAT

Per Reaction:

10 uL Dream Taq
0.3 uL fwd primer
0.3 uL rev primer
1 uL gDNA
8.4 uL H2O

Reaction Conditions:

-Hot Start 95C for 2:30min

32 Cycles (Only low/moderate yield, could bump up to 36)

-95C for 0.15

-50C for 0.25

-72C for 1.00

-72C for 1.00 before hold at 10C

Example gel

